

Thus Eqs. (29.2.44) and (29.2.45) yield

$$2\rho_{\text{VAC}} = (2\pi)^3 \int dm^2 H^{(1)}(m^2) . \quad (29.2.49)$$

But Eq. (29.2.43) tells us that $H^{(1)}(m^2)$ vanishes except perhaps at $m^2 = 0$, so

$$H^{(1)}(m^2) = 2(2\pi)^{-3} \delta(m^2) \rho_{\text{VAC}} . \quad (29.2.50)$$

Thus we see again that a non-vanishing vacuum energy density entails the existence of a massless fermion, the goldstino. Using Eq. (29.2.7), a straightforward calculation of the one-goldstino contribution to the spectral functions yields

$$H^{(1)}(m^2) = (2\pi)^{-3} \delta(m^2) |F|^2 . \quad (29.2.51)$$

Comparing this with Eq. (29.2.50) gives our previous result (29.2.8) for the vacuum energy density.

29.3 Non-Perturbative Corrections to the Superpotential

We saw in Section 27.6 that the superpotential in general supersymmetric theories of gauge and chiral superfields is not renormalized to any finite order of perturbation theory, so that if supersymmetry is not broken in the tree approximation then it can only be broken by non-perturbative corrections to the Wilsonian effective Lagrangian. We now take up a general analysis of these corrections. These were thoroughly studied in a series of papers by Affleck, Davis, Dine, and Seiberg in the early 1980s,³ with special attention to the case of supersymmetric versions of quantum chromodynamics with arbitrary numbers of colors and flavors. Here we will present a somewhat simplified analysis of general supersymmetric gauge theories, based on the more recent holomorphy arguments of Seiberg⁴ already used in Section 27.7.

To study non-perturbative effects, we shall here again consider a general renormalizable supersymmetric theory, but now including a possible θ -term in the Lagrangian density

$$\mathcal{L} = [\Phi^\dagger e^{-V} \Phi]_D + 2 \operatorname{Re} [f(\Phi)]_{\mathcal{F}} + \operatorname{Re} \left[\frac{\tau}{8\pi i} \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}} , \quad (29.3.1)$$

where the superpotential $f(\Phi)$ is a gauge-invariant cubic polynomial in the left-chiral superfields and τ is the parameter (27.3.23):

$$\tau = \frac{4\pi i}{g^2} + \frac{\theta}{2\pi} . \quad (29.3.2)$$